*4.4.1 Writers' subjective preferences toward personalized AI.* Comparing their experiences co-writing with the personalized AI and the non-personalized AI, writers reflected on both the positive and negative impact of personalization on authentic writing after we revealed that one of the tools they experienced was powered by an LLM personalized with the writing sample they provided. Overall, in our study, the majority of writers preferred working with personalized writing tools when they were asked to compare the two options. This is because writers believed personalization could help preserve their genuine voice, express themselves naturally, and better connect with their own identities.

In this regard, participants identified two main positive outcomes. First, many participants noted they produced better quality of work under time constraints when working with personalized AI; it reduced the need for going back and forth to adjust and align the generated text with their own voices and allowed participants to focus on producing new content. Generated output that was viewed as higher quality also yielded more inspiration for writers. As P15 suggested a stark contrast in their comment:

| Variable | Round 1 | | | | Round 1 (removed 7 responses) | | | | Round 2 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Opinions about AI writing assistance** | | | | | | | | | | | | |
| | $Z$ | | $p$ | | $Z$ | | $p$ | | $Z$ | | $p$ | |
| **Familiarity with generative AI** | 5.21 | | $< 0.001^{***}$ | | 5.63 | | $< 0.001^{***}$ | | 5.54 | | $< 0.001^{***}$ | |
| **Interest in reading AI-assisted writing** | 4.13 | | $< 0.001^{***}$ | | 5.47 | | $< 0.001^{***}$ | | 5.82 | | $< 0.001^{***}$ | |
| **Authenticity of AI-assisted writing** | 5.38 | | $< 0.001^{***}$ | | 2.58 | | $0.010^{*}$ | | 4.21 | | $< 0.001^{***}$ | |
| **Reading writing samples** | | | | | | | | | | | | |
| | $\beta$ | $S.E.$ | $t$ | $p$ | $\beta$ | $S.E.$ | $t$ | $p$ | $\beta$ | $S.E.$ | $t$ | $p$ |
| **Likeability** | | | | | | | | | | | | |
| Solo vs. Personalized AI | −0.25 | 0.20 | −1.26 | 0.215 | −0.15 | 0.16 | −0.90 | 0.371 | −0.24 | 0.15 | −1.56 | 0.125 |
| Solo vs. Non-personalized AI | −0.05 | 0.20 | −0.25 | 0.803 | −0.09 | 0.16 | −0.52 | 0.990 | −0.15 | 0.15 | −0.97 | 0.335 |
| **Enjoyment** | | | | | | | | | | | | |
| Solo vs. Personalized AI | −0.15 | 0.21 | −0.71 | 0.483 | −0.11 | 0.18 | −0.61 | 0.543 | −0.18 | 0.16 | −1.10 | 0.275 |
| Solo vs. Non-personalized AI | −0.15 | 0.21 | −0.70 | 0.483 | −0.04 | 0.18 | −0.20 | 0.839 | −0.12 | 0.16 | −0.74 | 0.465 |
| **Creativity** | | | | | | | | | | | | |
| Solo vs. Personalized AI | 0.91 | 0.25 | 0.01 | 0.999 | −0.11 | 0.22 | −0.50 | 0.619 | −0.18 | 0.19 | −0.92 | 0.360 |
| Solo vs. Non-personalized AI | −0.45 | 0.24 | −1.82 | 0.077 | −0.22 | 0.22 | −1.00 | 0.322 | −0.29 | 0.19 | −1.54 | 0.129 |
| **Comparing all writing samples** | | | | | | | | | | | | |
| | $\beta$ | $S.E.$ | $t$ | $p$ | $\beta$ | $S.E.$ | $t$ | $p$ | $\beta$ | $S.E.$ | $t$ | $p$ |
| **Likelihood of human writing** | | | | | | | | | | | | |
| Solo vs. Personalized AI | −0.79 | 0.35 | −2.25 | $0.028^{*}$ | −0.77 | 0.28 | −2.72 | $0.008^{**}$ | −0.73 | 0.25 | −2.94 | $0.004^{**}$ |
| Solo vs. Non-personalized AI | −0.32 | 0.35 | −0.90 | 0.372 | −0.54 | 0.28 | −1.90 | 0.061 | −0.64 | 0.25 | −2.57 | $0.012^{*}$ |
| **Comparing personalized vs. non-personalized writing samples** | | | | | | | | | | | | |
| | $\beta$ | $S.E.$ | $t$ | $p$ | $\beta$ | $S.E.$ | $t$ | $p$ | $\beta$ | $S.E.$ | $t$ | $p$ |
| **Preserving writers' authentic voices** | | | | | | | | | | | | |
| Personalized AI vs. Non-personalized AI | 0.40 | 0.27 | 1.47 | 0.164 | 0.05 | 0.24 | 0.19 | 0.853 | 0.07 | 0.21 | 0.36 | 0.722 |
| **Credits and authorship of work** | | | | | | | | | | | | |
| Personalized AI vs. Non-personalized AI | 0.55 | 0.25 | 2.24 | $0.037^{*}$ | 0.22 | 0.23 | 0.97 | 0.340 | 0.18 | 0.19 | 0.97 | 0.339 |
| **Opnions after reading AI-assisted writing** | | | | | | | | | | | | |
| | $Z$ | | $p$ | | $Z$ | | $p$ | | $Z$ | | $p$ | |
| **Perception of the writing** | 2.68 | | $0.007^{**}$ | | 2.68 | | $0.007^{**}$ | | 3.70 | | $< 0.001^{***}$ | |
| **Perception of the human writer** | 2.75 | | $0.006^{**}$ | | 3.41 | | $< 0.001^{***}$ | | 4.58 | | $< 0.001^{***}$ | |
| **Appreciation & evaluation of writing** | −0.39 | | 0.703 | | −0.25 | | 0.807 | | −0.30 | | 0.763 | |

Table 7. Results of statistical tests for numeric variables in Part 2 ($p^{*} < 0.05$, $p^{**} < 0.01$, $p^{***} < 0.001$)

> "*With the first one [working with personalized AI writing assistance] ... it almost seems like when you have a partner that energizes you and you get inspiration from each other. In the second round [working with non-personalized AI writing assistance], I felt like [...] almost like you have someone that gives a lot of suggestions that you don't really like, but it didn't really necessarily light me up.*"

In the same vein, some participants expressed a stronger sense of what they described as akin to *collaboration* when working with personalized AI. When working with a non-personalized AI, participants more often saw the need to switch to a supervising role to oversee and ensure the passage presented a consistent style. Such experiences were described as "a solo thing" (P05). By contrast, participants noted that it felt more like some type of "bilateral exchange" when they could focus on throwing out raw content—like the suggestions generated by the AI tool.

On the negative end, participants worried personalization might also lead to writers adopting more suggestions from AI, allowing more influences from the tool. Based on their subjective reflections, many participants believed that they contributed much more to the written content when working with a non-personalized AI, as they more frequently experienced the need to revise content generated by non-personalized AI, leaving more limited room for AI to influence them. In particular, more experienced writers were concerned that novice writers were more likely affected by frequently adopting suggestions from personalized AI, as they might not yet have established their own styles and voices in written work.

*4.4.2 Writers adopted a similar degree of assistance from both personalized and non-personalized AI tools.* Despite writers' subjective preferences for the personalized tool, in our study there was no significant difference in the frequency of requesting AI assistance when working with the personalized vs. the non-personalized tool ($\beta = 0.13$, $S.E. = 1.08$, $t = 0.12$, $p = 0.904$). Likewise, writers' behavioral data from the writing logs showed no significant difference in the rate of accepting suggestions from AI between the personalized vs. non-personalized condition ($\beta = -0.02$, $S.E. = 0.07$, $t = -0.27$, $p = 0.790$). (Also see Appendix D for detailed statistics for each writer.)

*4.4.3 Readers' responses to personalized and non-personalized AI-assisted writing.* The readers in the second part of our study reported similar experiences reading the three types of passage—writers' solo work, the work they produced with personalized AI, and the work with non-personalized AI. Per readers' numeric ratings, the degree of enjoyment ($F = 0.63$, $p = 0.536$), likeability ($F = 1.23$, $p = 0.298$), and creativity ($F = 1.19$, $p = 0.311$) after reading each type of passage showed no significant difference across the three conditions. Pair-wise comparison also showed no significant difference (See Table 7 for statistics). Furthermore, when asked to select portions of text they believed to be generated by AI, readers were not able to precisely identify which part of the text contained AI-assisted output. The rate of correct identification has no significant difference between the personalized and non-personalized conditions.

When asked to compare across the three writing passages, our reader participants rated the solo human work as more likely to be done independently by a human writer than the two forms of AI-assisted writing (solo − personalized AI: $\beta = -0.73$, $S.E. = 0.25$, $t = -2.94$, $p = 0.004$; solo − non-personalized AI: $\beta = -0.64$, $S.E. = 0.24$, $t = -2.57$, $p = 0.011$), while there is no significant difference between the work co-written with the personalized vs. non-personalized AI ($\beta = 0.09$, $S.E. = 0.26$, $t = 0.35$, $p = 0.727$). When comparing the two AI-assisted writing pieces to each writer's solo work, readers did not notice any difference regarding whether the AI-assisted work preserved the writer's authentic voice ($\beta = 0.07$, $S.E. = 0.21$, $t = 0.36$, $p = 0.722$). It is also worth noting that, on average, reader participants rated both versions of AI as "preserving the writer's authentic voice" from "moderate" (3 on a 5-point scale) to "a lot" (4 on a 5-point scale). Figures 3 and 4 also further illustrate the differences between participants' ratings for the conditions.

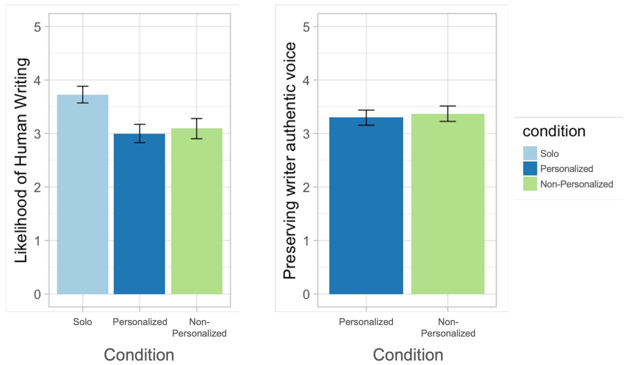


Fig. 3. Reader participants' ratings to compare the three pieces of writing. Left: Ratings for whether a piece of writing is likely to be done independently by a human writer (rating = 5) or co-written with AI (rating = 1) on a 5-point Likert scale. Right: Ratings for whether a piece of AI-assisted writing preserves a writer's authentic voice.

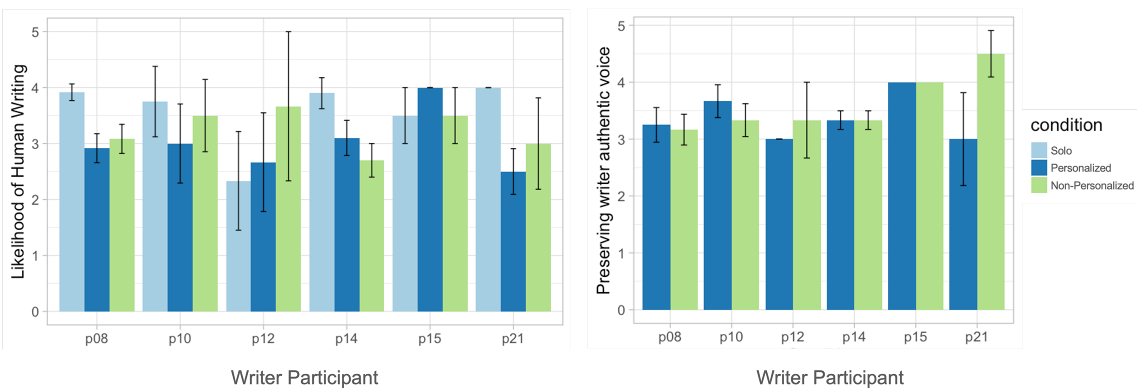


Fig. 4. Reader participants' ratings for writing passages for each writer. Left: Ratings for whether a piece of writing is likely to be done independently by a human writer (rating = 5) or co-written with AI (rating = 1) on a 5-point Likert scale. Right: Ratings for whether a piece of AI-assisted writing preserves a writer's authentic voice.

We took a closer look at readers' ratings for writing from each of the six writers respectively. For 4 out of the 6 writers, work co-written with non-personalized AI, compared to their work co-written with personalized AI, was perceived as more likely done by a human writer independently. The two writers (P14 and P15) whose work was rated as more likely to be written by a human when they worked with the personalized AI wrote about romance and spiritual topics, respectively. When we cross-checked the ratings from the readers with these writers' behavioral data, we also noticed that these two writers accepted more suggestions from the personalized AI than from the non-personalized during the writing sessions (P14 accepted 67% of suggestions from the personalized AI vs. 41% from the non-personalized AI; P15 accepted 71% suggestions from the personalized AI vs. 20% from the non-personalized AI).

We note that recent research has found that people could not reliably distinguish whether a piece of content is generated by people or the latest LLMs [33]. While our empirical results suggest that some readers might be able to tell if a piece is co-written with AI with some degree of confidence, they may not respond differently to content written by the writer alone or content co-written with AI in terms of how much they enjoy reading it, at least not in a "leisurely